UNIVERSITY OF BIRMINGHAM DUBAI

برنامج الذكاء الاصطناعي
THE ARTIFICIAL INTELLIGENCE PROGRAM

The Autonomous Road Inspection Agent

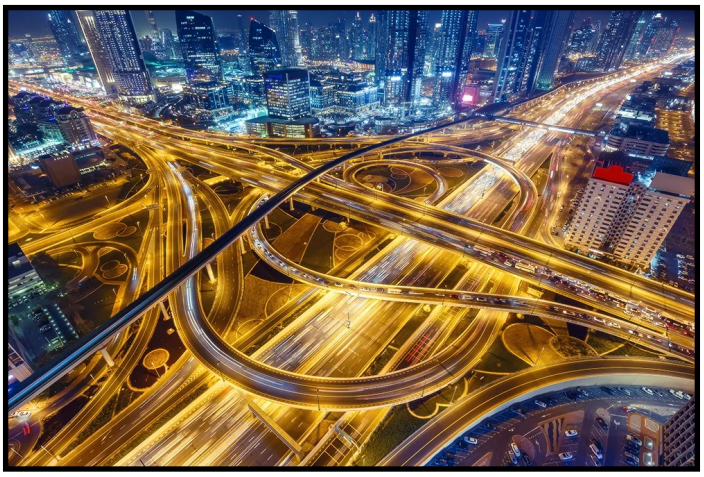
What information (data) does an AI agent need to monitor road conditions and identify damage?

The Mission

You are developing an AI agent to monitor and manage the road of Dubai.

Your agent is responsible for:
- Monitoring road conditions and identifying potential damage, such as potholes, cracks, and surface deterioration.
- Identifying roads that require further inspection.
- Recommending which roads should be inspected or repaired.
- Learning from previous observations and maintenance outcomes to improve future decisions.

It is up to you to design this system!



Section 1: What does the agent need to know?

This section is focused on thinking about what information your agent needs to operate on. For now, do not think about *preprocessing*, but just what *data sources* your agent needs.
For each data item, mention:
- **where** this information will come from
- **how** often it must be updated
- **why** the agent needs this data

*Think about what data your agent needs from each **road or road segment**. What information will the agent need to understand its current condition, identify potential damage, and determine whether further inspection is necessary? How would you **collect it**?*

*Think about what data your agent needs from the **surrounding road network and environment**. What information will the agent need to understand road usage, safety risks, and factors that may contribute to deterioration? How would you **collect it**?*

*Think about what data your agent needs from **each inspection or maintenance activity**. What information will the agent need to prioritise repairs, track completed work, and understand the effectiveness of previous interventions? How would you **collect it**?*

Section 2: From Raw Data to Actionable Information (Data Preprocessing and Representation)

Your agent now has access to the data sources you identified in Section 1 related to road conditions, the surrounding road network, and inspection and maintenance activities.However, raw data is not necessarily ready for an AI agent to use. Some observations may be incomplete, noisy, duplicated, outdated, recorded in different formats or geographical coordinate systems, or too large to process efficiently.

Design a preprocessing pipeline that transforms your collected data into reliable, compact, and useful information for the agent's decisions.

Consider the data sources you identified in Section 1. For three different data sources, answer the following questions:

1. What **data-quality problems** might you encounter? Consider missing values, noisy measurements, incorrect damage labels, duplicate reports, inconsistent road identifiers, and outliers.
2. What **preprocessing techniques** would you apply to address these problems? Be specific: would you filter noisy sensor measurements, interpolate missing values, remove duplicate reports, standardise geographical coordinates, correct inconsistent records, or flag suspicious observations?
3. When might preprocessing introduce **errors**? ***Think carefully***: Should all missing road-condition measurements be filled in? When is it better to preserve missing information and communicate uncertainty about a road's condition to the agent?

[Data Source 1] _____

[Data Source 2] _____

[Data Source 3] _____

If different sources need to be combined, how would you **align them**? For example, matching GPS coordinates from road images to the correct road segments.

Some of your chosen sources may contain **large amounts of information**, such as vehicle-mounted camera footage, satellite imagery, detailed geographical maps, or extensive inspection and maintenance records.

The agent cannot necessarily process or store all this information in its original form. Identify one or two data sources that would be particularly expensive to store, transmit, or process.

What information from these sources would actually be useful to the agent?

How would you transform the raw data into a smaller, structured representation? For example, could vehicle camera footage be converted into records of detected potholes, cracks, and their geographical locations? Could satellite imagery or detailed road maps be transformed into a structured representation of road segments, their characteristics, and surrounding infrastructure?

What information might be lost through compression, aggregation, or feature extraction? Under what circumstances would you need to retain or revisit the original data?

Section 3: Memory and Data During Deployment

Your autonomous road maintenance agent is now operational across Abu Dhabi's five districts. Every day, the system receives new road observations, citizen reports, inspection results, and maintenance updates, generating new information about road conditions, recurring defects, and the effectiveness of previous interventions.

Designing your Agent's Memory

What information should your agent remember about individual road segments, geographical areas, and previous inspection and maintenance activities to make better decisions in the future? Design three types of memory.

Design **three types of memory**. Specify **what information each contains**, why it would help the agent **make better decisions over time**, how it is **updated**, and when observations should expire or be revalidated.

[Memory 1] _____

[Memory 2] _____

[Memory 3] _____